

State Asymmetry in the In-In Effective Action

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

36. Vacuum, selector, and state functionals

The in-in effective action separates schematically as

$$\Gamma_{\text{CTP}} = \Gamma_{\text{vac}} + \Gamma_Q + \Gamma_\rho.$$

Exact microscopic Z_6 implies

$$\Gamma_{\text{vac}}[a] = \sum_{q \in \mathbb{Z}} c_{6q} e^{i6qa/f_a},$$

so

$$c_p^{\text{vac}} = 0, \quad p = 1, \dots, 5.$$

During a transient one-hot selector background, the action may contain

$$\epsilon^p e^{ipa/f_a} Q_{-p}[Q] + \text{h.c.},$$

where the selector carries compensating cyclic charge. These terms vanish when $Q \rightarrow 0$.

An asymmetric density matrix can also produce lower harmonics through occupation charges

$$\mathcal{N}_p = \sum_k n_k e^{-2\pi i p k / 6}.$$

These are intended finite-density forces. They dilute or disappear with the state. The central covariance relation is

$$\Gamma_{z\rho z^{-1}}[z\phi, zQ] = \Gamma_\rho[\phi, Q].$$

A forbidden lower harmonic must therefore be accompanied by selector or state charge. Under the displayed interaction graph, neither can become a permanent state-independent Wilson coefficient merely by disappearing.

37. Two-loop result

At two loops, the selector-reheaton-bath graph and the a -dependent threshold-QCD graph are not directly joined by a primitive vacuum vertex. Selector information reaches the threshold through propagators and occupation numbers prepared by reheating. Decomposing

$$S_k = S_{k,\text{vac}} + \delta S_{k,\rho}, \quad D_k = D_{k,\text{vac}} + \delta D_{k,\rho},$$

shows that the complete-orbit all-vacuum term cancels the forbidden harmonics; every $p < 6$ contribution contains at least one state insertion.

The physical equation takes the form

$$\square a + V'_0(a) + \sum_k M'_k(a) \langle \bar{\Psi}_k \Psi_k \rangle_{\rho, \text{ren}} = 0.$$

In influence-functional language, the environment supplies:

- a local finite-density force;
- a retarded memory and dissipation kernel;
- a noise kernel.

A decaying state can leave temporary memory but not automatically a permanent local vacuum term. A conserved relic or hydrodynamic zero mode would preserve a state-dependent source, not a new vacuum spurion.

The NLO thermal-QCD audit increased focusing by about 6.19% and shifted the selected phase by only 1.72×10^{-4} rad. The branch mechanism was perturbatively stable at this order [ThermalPressure2021].

38. Initial-state renormalisation

Non-vacuum initial states can introduce ultraviolet divergences. For UV-soft or Hadamard-like data, ordinary bulk counterterms suffice. More general short-distance structure requires counterterms on the initial hypersurface,

$$S_{\partial, \rho} = \int_{t=t_0} d^3x \mathcal{L}_{\partial, \rho}.$$

This is distinct from a permanent late-time bulk term. The preparation protocol must still be explicitly renormalised [CollinsHolman2005; Berges2005].

39. Necessary protection package

Exact Z_6 is necessary but not sufficient. The UV theory must also satisfy

$$\epsilon \rightarrow 0 \implies U(1)_a \text{ shift symmetry is restored.}$$

Every nonderivative a dependence must carry explicit powers of the same breaking spurion. The robust package is

exact Z_6 + continuous-shift spurion counting + no direct Q -threshold portal + UV-soft state preparation.